FACULTY OF COMPUTING

SEMESTER 1/20252026

SECJ 3553 - ARTIFICIAL INTELLIGENCE

SECTION 03

ASSIGNMENT 1

AI-Powered Intelligent Triage Assistant

THEME:

HEALTHCARE

LECTURER:

DR. NORSHAM BINTI IDRIS

Group no: Group 4

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Knowledge Representation

1. Patient Severity Prediction

AI analyses vital signs and symptoms to predict the severity level in real time, using logical inference, and it immediately identifies and assigns the highest triage level. If the patient's oxygen level is below 90% and body temperature exceeds 38 °c, and heart rate goes over 110 bpm, then the system will classify the patient's condition as critically severe.

• Knowledge Representation

IF oxygen_low = TRUE ∧ body_temp_high = TRUE ∧ heart_rate_high = TRUE, THEN severity_critical = TRUE

• Predicate Logic / First Order Logic:

lowOxygen(patient) ∧ highTemperature(patient) ∧ highHeartRate(patient) → criticalSeverity(system, patient)

Oxygen low	Body temp high	Heart rate high	Critical severity	(O_2 ∧ Temp ∧ HR → Critical)
T	T	T	T	T
T	T	F	F	T
T	F	T	F	T
F	T	T	F	T
F	F	F	F	T

2. Shock Alert System

If the patient's heart rate exceeds 120bpm and the blood pressure falls below 90/60, the system will trigger a shock alert to notify medical staff.

• Knowledge Representation

IF heart_rate_high = TRUE ∧ blood_pressure_low = TRUE, THEN alert_shock = TRUE

• Predicate Logic / First Order Logic:

highHeartRate(patient) ∧ lowBloodPressure(patient) → shockAlert(system, patient)

Heart rate high	Blood pressure low	Shock alert	(HR ∧ BP → Alert)
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

3. Symptom Pattern Recognition

If the system detects a symptom pattern that matches a known disease, then the system will suggest a diagnosis.

• Knowledge Representation

IF symptom_match = TRUE, THEN suggest_diagnosis = TRUE

• Predicate Logic / First-Order Logic

match(symptom_pattern, disease) → suggestDiagnosis(system, patient)

Symptom match	Suggest diagnosis	Symptom -> Diagnosis
T	T	T
T	F	F
F	T	T
F	F	T

4. Critical Alert System

If the patient's condition is deteriorating OR their heart rate becomes critical, then the system will alert medical staff.

• Knowledge Representation

IF condition_deteriorate = TRUE OR heart_rate_critical = TRUE, THEN alert_staff = TRUE

• Predicate Logic / First-Order Logic

deterioratingCondition(patient) ∨ criticalHeartRate(patient) → alertStaff(system, medical_team)

Condition deteriorate	Heart rate critical	Alert staff	(Cond ∨ HR → Alert)
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

5. Bed & Resource Allocation

Once a patient's severity is determined, AITA automatically checks bed availability. If a high-priority patient needs a bed and one is available, the system immediately assigns it, reducing waiting time and improving emergency response efficiency.

• **Knowledge Representation**

IF high triage_priority = TRUE AND bed_available = true THEN assign_bed = patient_id

• **Predicate Logic / First Order Logic**

∀patient ((highPriority(patient) ∧ ∃bed bedAvailability(bed)) → ∃bed assignBed(patient, bed))

High Priority	Bed Availability	Assign Bed	Priority ∧ BedAvail → Assign Bed
T	T	T	**T**
T	F	F	**T**
F	T	F	**T**
F	F	F	**T**

6. Predictive Equipment Maintenance

AITA monitors medical devices to ensure reliability by tracking usage hours and error rates. When a device exceeds its usage threshold or shows frequent errors, the system predicts potential failure and flags it for maintenance, preventing breakdowns and ensuring continuous patient care.

• **Knowledge Representation**

IF usage_hours > threshold OR error_rate = high, THEN maintenance_required = true

• **Predicate Logic / First Order Logic**

∀device ((usageHours(med_device) > threshold ∨ highErrorRate(med_device)) → maintenanceRequired(med_device))

Usage Hours	High Error Rate	Maintenance Required	Hours ∨ Error Rate → Maintenance Required
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

Knowledge Representation Involved to Achieve Goals

The primary objective of the triage system's Knowledge Representation (KR) is to enable AI to determine the severity of a patient's condition by using information from vital sign sensors, including body temperature, oxygen saturation, and heart rate. As soon as the patient's condition is identified, information is sent straight to the medical system's triage interface, which is reflected in the first KR. The system makes sure that patient data is updated continuously and in real-time in the second KR so that medical personnel are always aware of the most recent status.

When critical conditions are identified, the system will automatically send alerts to medical personnel based on the TRUE and FALSE logic rules specified in the KR. The rule is as follows: IF severity = TRUE, THEN send_alert = TRUE. Additionally, by recommending suitable treatment priorities and allocating hospital resources, like beds, according to patient urgency, the system will support medical staff.

Last but not least, the remaining KRs stand for the features of the integrated medical support system, which can identify patterns in symptoms to propose potential diagnoses and help medical personnel make decisions. In order to guarantee ongoing medical readiness and minimize treatment delays, the KR also makes sure that equipment conditions are tracked and alerts personnel when maintenance is necessary.

Peer review on Team Working (TW)

Category	4	3	2	1
Active participation	Routinely provides useful ideas when participating in the group.	Usually provides useful ideas when participating in the group.	Sometimes provides useful ideas when participating in the group.	Rarely provides useful ideas when participating in the group.

MEMBERS	ELIJAH SHE YU SHENG	LAU YEE WEN	CHERYL CHEONG KAH VOON	MUHAMMAD ADAM BIN RAZALI
ELIJAH SHE YU SHENG	-	4	4	4
LAU YEE WEN	4	-	4	4
CHERYL CHEONG KAH VOON	4	4	-	4
MUHAMMAD ADAM BIN RAZALI	4	4	4	-